AdifySkills

n8n Automation Mastery
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CLASS NOTES

Module 6 · Production-Grade Engineering

DAY 19 of 25

Self-Hosting n8n: Docker, VPS & Domains

"Apna khud ka n8n server — unlimited workflows."

⚡ AAJ HUM YEH BANAYENGE

Self-Hosted n8n + Uptime Monitor

Aaj VPS par Docker Compose se n8n + Postgres + Caddy (auto HTTPS) chalate hain — ready files neeche hain. Saath mein ek chhota workflow jo har 5 minute aapke n8n ka /healthz check karta hai aur down hone par Telegram alert deta hai.

Docker | Docker Compose | Postgres | Nginx/Caddy

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Self-hosting = unlimited executions, data apne paas, client ko kam cost. Har serious automator ko aana chahiye.

🎯 Aaj ke baad aap yeh kar paoge

01 Cloud vs self-hosted: cost & control

🌍 Kiraya vs EMI.
⚙️ Cloud: easy, limits. Self-host: sasta, full control, maintenance.

02 Docker & Docker Compose basics

🌍 Tiffin box — sab ek dabbe mein.
⚙️ docker-compose.yml: n8n + postgres services, volumes.

03 VPS setup (Hostinger VPS / any Ubuntu server)

🌍 Khaali plot.
⚙️ Ubuntu VPS → SSH → Docker install → firewall.

04 Custom domain + SSL (reverse proxy)

🌍 Ghar ka naam board + taala.
⚙️ DNS A record → Caddy/Nginx reverse proxy → Let's Encrypt SSL.

05 Environment variables & Postgres backend

🌍 Ghar ki settings ek list mein.
⚙️ N8N_HOST, WEBHOOK_URL, N8N_ENCRYPTION_KEY, DB_TYPE=postgresdb.

06 Updates, backups & restore

🌍 Ghar ki duplicate chaabi.
⚙️ pg_dump nightly + encryption key safe; docker compose pull && up -d.

⏱ Class ka plan (2 ghante)

10m	30m	55m	15m	10m

`0:00-0:10` **Recap**

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

`0:10-0:40` **Concept: Hosting Architecture**

🪄 Kiraye ke ghar mein maalik ke rules. Apne ghar mein tumhare rules — par safai bhi tumhe karni hai.

`0:40-1:35` **Live Setup**

Fresh VPS → Docker → n8n + Postgres → domain + SSL

🪄 Aaj ki kahani yaad hai — 'Kiraye ka flat vs apna ghar'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

`1:35-1:50` **Checklist Review**

Security hardening

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

`1:50-2:00` **Q&A + Homework**

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

VPS

Cloud mein apna virtual computer (Ubuntu).

🪄 **Analogy:** Kiraye ka flat jahan aap kuch bhi rakh sakte ho

CONCEPT 02

Docker

App + uski saari dependencies ek "container" mein.

🪄 **Analogy:** Tiffin box — sab packed

CONCEPT 03

Reverse proxy + SSL

Domain → n8n, HTTPS ke saath (Caddy/Nginx).

🪄 **Analogy:** Building ka reception

CONCEPT 04

Encryption key

Credentials encrypt karne ki key — kho gayi to credentials gaye.

🪄 **Analogy:** Locker ki master key

🔑 Aaj ka main idea

Self-hosting = n8n ko apne server (VPS) par Docker se chalana. Domain + HTTPS lagao, Postgres database use karo, aur regular backup lo.

- Hostinger VPS par n8n + Postgres
- n8n.meracompany.com HTTPS ke saath
- Har raat database backup Drive par

📖 Ek kahani se samjho: Kiraye ka flat vs apna ghar

1. 🏢 **Cloud = kiraye ka flat** = n8n Cloud
2. 🏠 **Apna plot liya** = VPS server
3. 📦 **Ready-made ghar ka dabba** = Docker container
4. 🪧 **Ghar ka address board** = Domain + SSL
5. 🗄️ **Tijori aur backup chaabi** = Postgres + backups

🪄 Ek line mein: “Kiraye ke ghar mein maalik ke rules. Apne ghar mein tumhare rules — par safai bhi tumhe karni hai.”

Gehraai se samjho — har concept andar se

Kya hai, n8n andar kya karta hai, kab use karein, kab nahi, example aur common galti.

01 VPS

📖 **Kya hai:** VPS (Virtual Private Server) cloud mein kiraye ka ek computer hai — usually Ubuntu Linux — jo 24×7 chalta hai. Ispar aap n8n install karke apna server chalate ho.

⚙️ **Andar kaise:** Aap provider (Hostinger, DigitalOcean, Hetzner, AWS Lightsail) se plan lete ho: CPU, RAM, disk. Ek public IP milta hai. SSH se terminal mein login karte ho (ssh root@IP). n8n ke liye shuru mein 2 vCPU + 4 GB RAM kaafi hai (₹500–900/month). Server ki updates, firewall aur backup aapki zimmedari.

✅ **Kab use karein**
Bahut executions, data apne server par rakhna, multiple clients ke workflows, ya n8n Cloud ka kharcha bachana.

⛔ **Kab nahi**
Agar aap/client server sambhalna nahi chahte aur executions kam hain — n8n Cloud zyada aasaan aur safe hai.

🧪 **Example:** Ubuntu 24.04 VPS → ssh root@203.0.113.10 → apt update && apt upgrade -y → Docker install.

⚠️ **Common galti:** Root password se login chalu chhod dena aur firewall na lagana — bots din mein hazaaron baar try karte hain. SSH key + ufw (22, 80, 443) lagao.

🪄 **VPS = kiraye ka flat. Aazaadi bhi, zimmedari bhi.**

02 Docker

📖 **Kya hai:** Docker app ko uski saari zaroori cheezon ke saath ek "container" mein pack karta hai — jaise tiffin box. Har server par same tarah chalta hai.

⚙️ **Andar kaise:** docker-compose.yml ek file hai jisme likhte ho kaunse containers chahiye (n8n, postgres, caddy), unke settings (.env se), aur data kahan save ho (volumes). docker compose up -d se sab chal jaate hain. Update: docker compose pull && docker compose up -d. Volumes ki wajah se container hatane par bhi data bacha rehta hai.

✅ **Kab use karein**
Har self-hosted n8n setup — official tareeka yahi hai. Update aur migration bahut aasaan.

⛔ **Kab nahi**
Volume ke bina n8n chalana — container restart hote hi saare workflows aur credentials gayab!

🧪 **Example:** services: n8n (image: n8nio/n8n), postgres (image: postgres:16), caddy (image: caddy:2) — teeno ek file mein.

⚠️ **Common galti:** "latest" tag par bina dekhe update kar dena — breaking change aaya to sab ruk gaya. Specific version pin karo, release notes padh kar update.

🪄 **Docker = tiffin box. Volume = data ka locker.**

03 Reverse proxy + SSL

📖 **Kya hai:** Reverse proxy (Caddy ya Nginx) domain (n8n.yourdomain.com) par aane wali requests ko andar n8n tak pahunchata hai, aur HTTPS (SSL certificate) sambhalta hai.

⚙️ **Andar kaise:** Step 1: DNS mein A record — n8n.yourdomain.com → VPS ka IP. Step 2: Caddyfile mein 2 line: domain { reverse_proxy n8n:5678 }. Caddy khud Let's Encrypt se free SSL certificate le leta hai aur renew karta hai. n8n ko batana padta hai uska public URL: N8N_HOST, N8N_PROTOCOL=https, WEBHOOK_URL=https://n8n.yourdomain.com/.

✅ **Kab use karein**
Har live setup. Webhooks, OAuth (Google login) aur Telegram trigger sab HTTPS maangte hain.

⛔ **Kab nahi**
n8n ka port 5678 seedha internet par khula mat chhodo — sab traffic proxy se aaye.

🧪 **Example:** Caddyfile: n8n.adifyskills.com { reverse_proxy n8n:5678 }

⚠️ **Common galti:** WEBHOOK_URL set na karna — webhook nodes mein localhost:5678 wale URL dikhte hain aur bahar se koi call nahi kar pata.

🪄 **Proxy = reception. DNS → Caddy → n8n, HTTPS free.**

04 Encryption key

📖 **Kya hai:** N8N_ENCRYPTION_KEY woh master key hai jisse n8n saare credentials (API keys, passwords, tokens) database mein encrypt karta hai.

⚙️ **Andar kaise:** Pehli baar chalne par n8n khud ek key bana leta hai (agar aapne nahi di). Credentials sirf usi key se khulte hain. Server migrate kiya, naya container banaya, ya database restore kiya — aur key alag hui — to saare credentials bekaar: "Credentials could not be decrypted".

✅ **Kab use karein**
Setup ke pehle din: .env mein khud lambi random key daalo aur password manager mein save karo.

⛔ **Kab nahi**
Key ko Git repo, screenshot ya WhatsApp par kabhi mat daalo — key + database = saare client credentials.

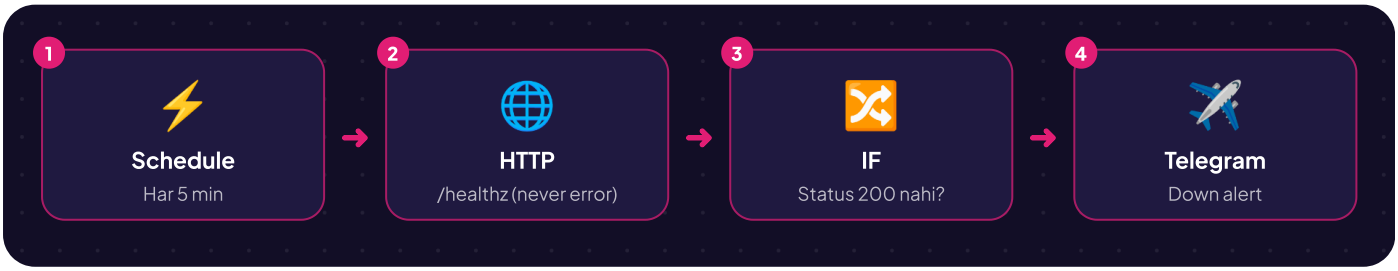
🧪 **Example:** .env: N8N_ENCRYPTION_KEY=8f3c…(32+ random characters) — openssl rand -hex 32 se banao.

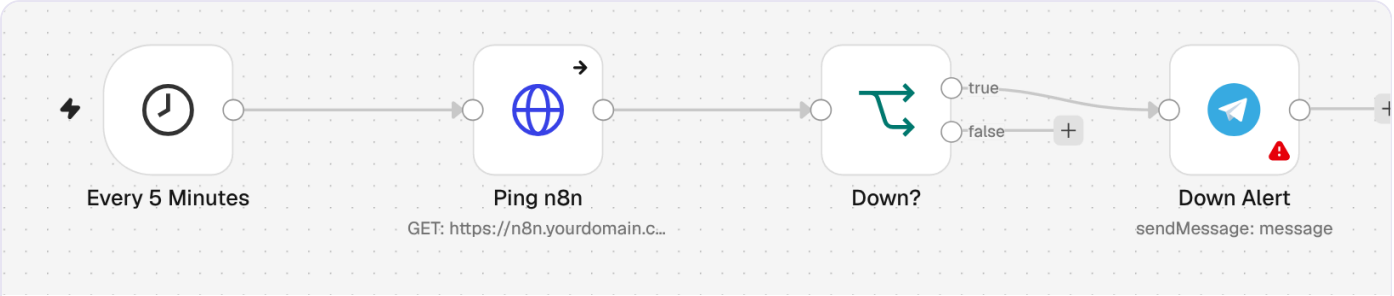
⚠️ **Common galti:** Backup mein sirf database rakha, key nahi — restore karne par saare 40 credentials dobara banane pade.

🪄 **Encryption key = locker ki master chaabi. Database ke saath key bhi backup.**

🗺️ Aaj ka workflow: Self-Hosted n8n + Uptime Monitor

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

Clients jo apna n8n chahte hain (VPS setup + monitoring) — aam taur par **₹8,000 – ₹20,000 + maintenance**.
(Andaaza hai; client aur scope ke hisaab se badalta hai.)

🛠 Step-by-step build — n8n kholo aur saath chalo

1. 🖥️ **SSH**
   VPS mein login, apt update
2. 🖥️ **SSH**
   Docker install: curl -fsSL https://get.docker.com | sh
3. 📄 **Read/Write Files from Disk**
   docker-compose.yml: n8n + postgres
4. 🖥️ **SSH**
   DNS A record → VPS IP; Caddy reverse proxy
5. 🖥️ **SSH**
   docker compose up -d
6. ⚡ **Schedule Trigger**
   Nightly backup workflow

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Telegram bot token (BotFather se)**
- [ ] URL mein apna domain daalo
- [ ] Best practice: yeh monitor kisi DOOSRE n8n / server par chalao (same server down hua to alert kaun bhejega?)

✅ Test kaise karein

1. Galat URL daal ke test karo → alert aana chahiye

Data ka safar — har node ke baad

Aaj ke build mein data kaise badalta hai. Expressions samajhne ki chaabi yahi hai.

1 🔧 Every 5 Minutes

Har 5 minute ek item.


```
{
  "timestamp": "2026-12-09T02:35:00.004+05:30",
  "Timezone": "Asia/Kolkata (UTC+05:30)"
}
```


2 ♾️ Ping n8n

Full response ON — body ke saath statusCode aur headers bhi. (Yahan server down hai: 502 Bad Gateway.)


```
{
  "statusCode": 502,
  "statusMessage": "Bad Gateway",
  "headers": {
    "server": "Caddy",
    "content-length": "0"
  },
  "body": ""
}
```


3 🔧 Down?

502 ≠ 200 → true → alert.


```
{
  "true (1 item)": "statusCode 502",
  "false (0)": "—"
}
```


4 🔧 Down Alert

Telegram par turant alert.

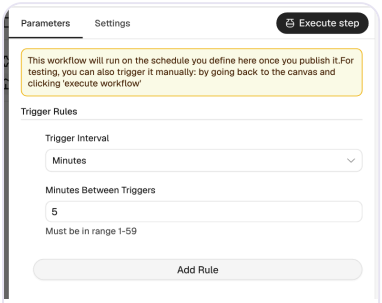

```
{
  "ok": true,
  "result": {
    "message_id": 951,
    "text": "🔴 n8n DOWN!\nStatus: 502\nTime: 02:35"
  }
}
```

Node anatomy — har setting ka matlab

Har zaroori option kya karta hai aur kab badalna hai. Isse aap khud naye workflows bana paoge.

Schedule Trigger — minutes

Har 5 minute monitor chalata hai.

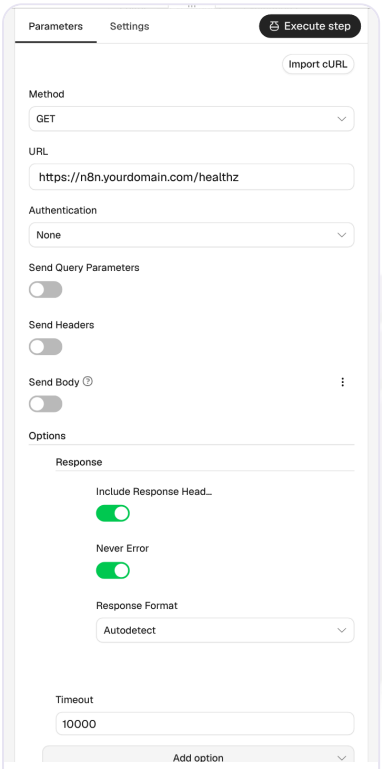


📸 Asli n8n: Schedule Trigger — minutes

Setting	Kya karti hai	Aaj ki value	Kab badlein
Trigger Interval	Minutes	`Minutes`	—
Minutes Between Triggers	Kitne minute ke gap par	`5`	1 minute = 1440 executions/din — zaroorat se zyada
Executions ka dhyan	Har run ek execution	`288/din`	Settings → Save successful executions: off (DB halka)

HTTP Request — health check

n8n ka /healthz URL check karta hai.

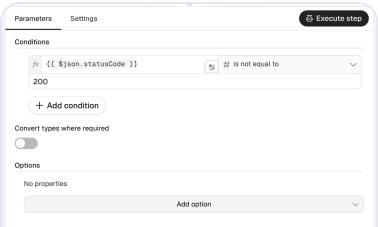


📸 Asli n8n: HTTP Request — health check

Setting	Kya karti hai	Aaj ki value	Kab badlein
URL	Aapke n8n ka health endpoint	`https://n8n.yourdomain.com/healthz`	Jawab: {"status": "ok"}
Options → Timeout	Kitni der rukna	`10000 ms`	10 sec mein jawab nahi = problem
Options → Response → Include Response Headers and Status	statusCode bhi chahiye	`On (fullResponse)`	Iske bina status code nahi milta
Options → Response → Never Error	404/500 par bhi node fail na ho	`On`	Taaki IF status dekh sake
Settings → On Error	Network fail (server band) par bhi aage	`Continue (regular output)`	Warna server down par yahin ruk jaata

🔀 IF — status check

200 nahi to alert.



📸 Asli n8n: IF — status check

Setting	Kya karti hai	Aaj ki value	Kab badlein
Value	{{ $json.statusCode }}	`200 / 502 / undefined`	Server band ho to statusCode hota hi nahi
Operator	Number → is not equal to	`200`	undefined ≠ 200 → true → alert
false branch	Sab theek — kuch mat karo	`khaali`	Daily "all OK" report alag workflow se

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

Cloud plan ₹2000+/mahina, limits

➜

⚡ Ab (n8n)

VPS ~₹500/mahina, unlimited workflows

⏱ Kharcha kam, control poora — lekin maintenance tumhari.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    environment:
      - N8N_HOST=n8n.yourbrand.com
      - WEBHOOK_URL=https://n8n.yourbrand.com/
      - DB_TYPE=postgresdb
      - N8N_ENCRYPTION_KEY=${N8N_ENCRYPTION_KEY}
    volumes: [n8n_data:/home/node/.n8n]
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "domain": "n8n.yourbrand.com",
  "vps": "2 vCPU / 4 GB RAM"
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "url": "https://n8n.yourbrand.com",
  "ssl": "valid",
  "db": "postgres"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude

📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Aaj VPS par Docker Compose se n8n + Postgres + Caddy (auto HTTPS) chalate hain — ready files neeche hain. Saath mein ek chhota workflow jo har 5 minute aapke n8n ka /healthz check karta hai aur down hone par Telegram alert deta hai.

Flow: Schedule (Har 5 min) → HTTP (/healthz (never error)) → IF (Status 200 nahi?) → Telegram (Down alert)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt

📋 Select karke copy karo

```
I'm building the n8n workflow "Self-Hosted n8n + Uptime Monitor" (nodes: Every 5 Minutes, Ping n8n, Down?, Down Alert).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
4 nodes · ready to import
`adifyskills.com/workflows/day-19-uptime-monitor.json` ↗

🌐 Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/19` ↗

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.

⚠️ Yeh galtiyan sabse common hain

! WEBHOOK_URL set na karna — webhooks localhost dikhate hain
! Encryption key backup na karna
! Volume mount bhool jaana — restart par sab data gayab

⭐ Pro tips — time bachane wali tricks

★ Hostinger VPS par n8n ka 1-click template bhi milta hai
★ Updates se pehle backup + release notes padho
★ Firewall: sirf 22, 80, 443 open

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Live: ek fresh Ubuntu VPS par n8n + Postgres Docker Compose ke saath, apne domain par HTTPS.

📝 Homework

Apna self-hosted instance setup karo aur ek workflow export/import karke migrate karo.

☐ Homework ho gaya

BEGINNER
Local Docker par n8n chalao ☐

INTERMEDIATE
VPS + domain + SSL ☐

ADVANCED
Automatic backups to S3/Drive ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Monitor ko 2 baar lagataar fail hone par hi alert bhejne do (false alarm kam).

💡 Hint 1
Ek baar fail hone par thoda ruk ke dobara check karo.

💡 Hint 2
Down? true → Wait 1 minute → Ping dobara → IF phir bhi down → alert.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🧪 Expression practice sheet

Input dekho aur khaali jagah mein expression likho. Jawab aakhri page par hain.

1. Status code

```
{"statusCode":200}
```

Mera jawab: _____

2. Down hai?

```
{"statusCode":502}
```

Mera jawab: _____

3. Jawab hi nahi aaya to

Input ki zaroorat nahi

Mera jawab: _____

4. /healthz ka body

```
{"body":{"status":"ok"}}
```

Mera jawab: _____

5. Server header (kaun jawab de raha)

```
{"headers":{"server":"Caddy"}}
```

Mera jawab: _____

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Webhooks ka public URL batane wala env?

- (A) NODE_ENV
- (B) WEBHOOK_URL
- (C) PORT
- (D) TZ

Q2. Credentials encrypt karne wali key?

- (A) SSH key
- (B) SSL cert
- (C) N8N_ENCRYPTION_KEY
- (D) API_KEY

Q3. Data restart ke baad bhi rahe — kya chahiye?

- (A) Cache
- (B) Cookie
- (C) Nothing
- (D) Docker volume

Q4. HTTPS ke liye?

- (A) Reverse proxy + SSL cert
- (B) Bada server
- (C) Postgres
- (D) Redis

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Docker
Software ko ek ready dabbe (container) mein chalane ka tool — har jagah same chalta hai.

VPS
Kiraye ka online computer/server jo 24×7 chalta hai.

SSL
Website ka taala — https wala lock, data safe rehta hai.

Schedule
Fixed time par workflow chalana — roz, har ghante, har Monday.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 20

Scaling, Security & Workflow DevOps

"Ek client se 50 clients tak — bina tootey."

☐ **Day 19 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
WEBHOOK_URL

Q2 → C
N8N_ENCRYPTION_KEY

Q3 → D
Docker volume

Q4 → A
Reverse proxy + SSL cert

✅ Challenge ka solution

Down? (true) → Wait 1 min → HTTP Ping n8n (copy) → IF statusCode != 200 → Down Alert.

🧪 Expression sheet ke jawab

1. `{{ $json.statusCode }}` → **200**
2. `{{ $json.statusCode !== 200 }}` → **true**
3. `{{ $json.statusCode || "no response" }}` → **no response**
4. `{{ $json.body.status }}` → **ok**
5. `{{ $json.headers.server }}` → **Caddy**

📚 Official docs

- Hosting n8n ↗
- Docker install ↗
- Environment variables ↗
- Server setups ↗

▶ Videos

- n8n self host docker compose ↗
- n8n hostinger vps install ↗

Doubt poochho
WhatsApp: +91 83838 91716
wa.me/918383891716

Day 19 lesson page
Quiz, copy buttons, videos
adifyskills.com/day/19

+91 83838 91716
Call / WhatsApp — Adify Skills